

Matrix Algebra

Part 3/3: Eigenvectors, Kernels and Matrix Transformations

CSC 1105 Mathematics for Computer Science

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Lecture Outline

- 2D and 3D transformations using matrices
- Eigenvalues and Eigenvectors
- Kernels
- Application: Low-rank image representation

Learning Outcomes

By the end of this lecture, you should be able to:

- 1 Represent a linear transformation with a matrix
- 2 Compute and compose common 2D and 3D transformations
- 3 Find eigenvalues and eigenvectors of a 2×2 matrix
- 4 Diagonalize a matrix when it has enough eigenvectors
- 5 Compute a basis for the kernel of a matrix
- 6 Apply rank-nullity to explain lost information
- 7 Relate eigen-computations to low-rank image approximation

Linear Transformations

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear when

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$$

and

$$T(c\mathbf{u}) = cT(\mathbf{u}).$$

Every linear transformation between finite-dimensional coordinate spaces has a matrix representation:

$$T(\mathbf{x}) = A\mathbf{x}, \quad A \in \mathbb{R}^{m \times n}.$$

The columns of A are the transformed standard basis vectors.

A Matrix Transformation Computed Directly

Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

Then

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2(3) + 1(-1) \\ 0(3) + 1(-1) \end{bmatrix} = \boxed{\begin{bmatrix} 5 \\ -1 \end{bmatrix}}.$$

Common 2D Transformation Matrices

Scaling

$$S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

Rotation

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Reflection in the x -axis

$$F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Horizontal shear

$$H = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

All four transformations are linear and leave the origin fixed:

$$A \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Translation with Homogeneous Coordinates

Translation by $\mathbf{t} = (t_x, t_y)^T$ maps

$$T(\mathbf{x}) = \mathbf{x} + \mathbf{t}.$$

If $\mathbf{t} \neq \mathbf{0}$, then $T(\mathbf{0}) = \mathbf{t} \neq \mathbf{0}$, so translation is **affine, not linear** in $\mathbb{R}^2$.

Represent (x, y) using homogeneous coordinates $(x, y, 1)^T$:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}.$$

For $(t_x, t_y) = (4, -2)$ and $(x, y) = (3, 1)$:

$$\begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ 1 \end{bmatrix}, \quad (3, 1) \mapsto (7, -1).$$

Key distinction: The 3×3 matrix defines a linear map in $\mathbb{R}^3$ that represents an affine translation in $\mathbb{R}^2$.

Order of Transformation Composition

Start with the point $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and apply:

$$\underbrace{S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}}_{\text{Double the horizontal coordinate}}, \quad \underbrace{R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}_{\text{Rotate } 90^\circ \text{ anticlockwise}}.$$

1. Scale first, then rotate:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{S} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \xrightarrow{R} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow RS\mathbf{x} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

2. Rotate first, then scale:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{R} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \xrightarrow{S} \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow SR\mathbf{x} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Order Matters

The same transformations in different orders can give different results: $RS \neq SR$ here. In a matrix product, the **rightmost matrix acts first**: $RS\mathbf{x} = R(S\mathbf{x})$.

A 3D Rotation About the z-Axis

The rotation matrix is

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For $\theta = 90^\circ$ and $\mathbf{p} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$:

$$R_z(90^\circ)\mathbf{p} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

The z-coordinate stays fixed while the xy-coordinates rotate.

Eigenvectors and Invariant Directions

A nonzero vector $\mathbf{v}$ is an eigenvector of A when

$$A\mathbf{v} = \lambda\mathbf{v}$$

for some scalar λ .

Interpretation

The transformation keeps the direction of $\mathbf{v}$ on the same line. The eigenvalue λ controls the scale and possible reversal.

- $|\lambda| > 1$ stretches that direction
- $0 < |\lambda| < 1$ contracts it
- $\lambda < 0$ reverses orientation on the line
- $\lambda = 0$ sends the eigenvector to the zero vector

The Characteristic Equation

From

$$A\mathbf{v} = \lambda\mathbf{v},$$

we obtain

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

A nonzero solution exists only when $A - \lambda I$ is singular:

$$\boxed{\det(A - \lambda I) = 0}.$$

This polynomial equation produces the eigenvalues. For each eigenvalue, solve

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

to find the corresponding eigenvectors.

Eigenvalues of a 2×2 Matrix

Let

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$

Then

$$\begin{aligned} \det(A - \lambda I) &= \begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix} \\ &= (4 - \lambda)(3 - \lambda) - 2 \\ &= \lambda^2 - 7\lambda + 10 \\ &= (\lambda - 5)(\lambda - 2). \end{aligned}$$

Therefore

$$\boxed{\lambda_1 = 5, \quad \lambda_2 = 2}.$$

Eigenvectors for $\lambda = 5$ and $\lambda = 2$

For $\lambda = 5$:

$$(A - 5I)\mathbf{v} = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0}$$

gives $y = x$. One eigenvector is

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For $\lambda = 2$:

$$(A - 2I)\mathbf{v} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0}$$

gives $y = -2x$. One eigenvector is

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Verification of the Eigenpairs

For $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$A\mathbf{v}_1 = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5\mathbf{v}_1.$$

For $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$:

$$A\mathbf{v}_2 = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2\mathbf{v}_2.$$

Every nonzero scalar multiple of an eigenvector belongs to the same eigenspace.

Diagonalization

Place independent eigenvectors in the columns of P and their eigenvalues on the diagonal of D :

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}.$$

Then

$$AP = PD$$

and

$$\boxed{A = PDP^{-1}}.$$

The matrix P changes coordinates to an eigenvector basis. In that basis, the transformation only scales coordinates.

Repeated Transformations with Diagonalization

From $A = PDP^{-1}$:

$$A^n = PD^nP^{-1} = P \begin{bmatrix} 5^n & 0 \\ 0 & 2^n \end{bmatrix} P^{-1}.$$

For $n = 4$:

$$A^4 = \begin{bmatrix} 422 & 203 \\ 406 & 219 \end{bmatrix}.$$

Applied to $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$A^4\mathbf{x}_0 = \begin{bmatrix} 422 \\ 406 \end{bmatrix}.$$

The component in the $\lambda = 5$ direction grows faster, so repeated multiplication increasingly aligns the result with $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

A Matrix That Cannot Be Diagonalized

Consider

$$C = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}.$$

Its characteristic polynomial is

$$\det(C - \lambda I) = (3 - \lambda)^2,$$

so $\lambda = 3$ has algebraic multiplicity 2.

However,

$$(C - 3I)\mathbf{v} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{0}$$

forces $y = 0$. The eigenspace has only one independent direction:

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

Two dimensions require two independent eigenvectors for diagonalization.

Kernel and Null Space

For $A \in \mathbb{R}^{m \times n}$, the kernel is

$$\ker(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}.$$

The kernel contains every input that the transformation sends to zero.

Two uses of the word kernel

In linear algebra, a kernel means a null space. In image convolution, a kernel means a small filter matrix. The meanings are related to different contexts.

The kernel of a linear transformation always forms a subspace of the input space.

Computing a Basis for a Kernel

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

The system $A\mathbf{x} = \mathbf{0}$ reduces to

$$x_1 + 2x_2 + 3x_3 = 0.$$

Set $x_2 = s$ and $x_3 = t$. Then

$$\mathbf{x} = \begin{bmatrix} -2s - 3t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore

$$\ker(A) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Rank and Nullity

For the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix},$$

the second row is twice the first, so $\text{rank}(A) = 1$.

The kernel has two basis vectors, so $\text{nullity}(A) = 2$.

Rank-nullity theorem

For $A \in \mathbb{R}^{m \times n}$,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Here,

$$1 + 2 = 3.$$

All Solutions from One Particular Solution

Suppose $A\mathbf{x} = \mathbf{b}$ has one solution $\mathbf{x}_p$. If $\mathbf{z} \in \ker(A)$, then

$$A(\mathbf{x}_p + \mathbf{z}) = A\mathbf{x}_p + A\mathbf{z} = \mathbf{b}.$$

For

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

one particular solution is $\mathbf{x}_p = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Hence every solution is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}.$$

Invertibility Revisited

For a square matrix $A \in \mathbb{R}^{n \times n}$, the following statements are equivalent:

- A^{-1} exists
- $\det(A) \neq 0$
- $\text{rank}(A) = n$
- $\ker(A) = \{\mathbf{0}\}$
- Zero is not an eigenvalue of A

Why zero matters

If $A\mathbf{v} = \mathbf{0}$ for a nonzero $\mathbf{v}$, then $A\mathbf{v} = \mathbf{0}$ and the kernel is nontrivial.

Geometric Transformations of a Banknote Image

Original



90° rotation



Horizontal reflection



Coordinate Transforms and Intensity Transforms

Image processing uses matrices in two different ways.

Pixel coordinates

A geometric matrix transforms positions:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix}.$$

Software then samples the image at the transformed coordinates.

Interpolation matters when transformed coordinates fall between pixels.

Pixel intensities

Flatten an $H \times W$ image into

$$\mathbf{x} \in \mathbb{R}^{HW}.$$

A linear image operation has the form

$$\mathbf{y} = M\mathbf{x}.$$

Singular Value Decomposition

For a grayscale image matrix $Y \in \mathbb{R}^{H \times W}$:

$$Y = U\Sigma V^T.$$

The diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \dots \geq 0$$

are singular values.

The right singular vectors are eigenvectors of $Y^T Y$:

$$Y^T Y \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i.$$

Keeping the first k singular values gives the rank- k approximation

$$Y_k = U_k \Sigma_k V_k^T.$$

Low-Rank Approximations of the Banknote Matrix

Rank 5



Rank 20



Rank 50



Higher rank restores finer texture and sharper edges.

Revision Questions

- 1 Find the eigenvalues and eigenvectors of

$$B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

- 2 Determine whether B can be diagonalized.
- 3 Find a basis for the kernel of

$$C = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \end{bmatrix}.$$

- 4 Verify rank-nullity for C .

Summary

- Matrices represent linear transformations in coordinates
- Matrix multiplication composes transformations in right-to-left order
- Eigenvectors identify directions preserved by a transformation
- Diagonalization simplifies repeated matrix powers
- The kernel records inputs that a transformation loses
- Rank-nullity divides the input dimensions into retained and lost directions
- Eigenvalue computations can be used to construct low-rank approximations of images

Questions?